

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (CE/IT/EE)

December 2011

CL 102.01 Environmental Sciences

Date: 14.12.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Discuss relationship between Man and Environment. [03]
(b) Write a short note on: Lithosphere. [04]
- Q - 2 (a) What are causes of Environmental Pollution? List all Environmental Pollution and their sources in detail. [04]
(b) Define Noise Pollution. Describe all effects of Noise pollution. [05]
(c) Explain objective & guiding principles of Environmental Studies. [05]

OR

- Q - 2 (a) Give general classification of pollutants and explain classification of pollutants based On its origin. [04]
(b) Give the term Environmental degradation and list out all possible types of Environmental pollution. [05]
(c) Discuss the temperature profile of atmosphere in detail. [05]
- Q - 3 (a) Explain effects of air pollution on human being. [04]
(b) Describe sources and effects of water pollutants. [05]
(c) Describe the main causes of depletion of ozone layer and briefly comment over control measures taken to prevent further ozone depletion at international level. [05]

OR

- Q - 3 (a) Give classification of water pollutants and describe organic water pollutants. [04]
(b) Enlist all current global environmental problems and discuss greenhouse effect in details. [05]
(c) Define Eutrophication, its causes and effects. [05]

SECTION – II

- Q - 4 (a) Define 'Ecosystem'. Write objectives of ecological studies. [03]
(b) What do you understand by structure of ecosystem? Give classification of consumers. [04]
- Q - 5 (a) Explain the concept of Food Chain with example. [04]
(b) Write environmental significance of nitrogen cycle and Hydrological cycle. [05]
(c) Define 'Food Web'. Explain Ecological Pyramid of Numbers. [05]

OR

- Q - 5 (a) Discuss how Energy Flow differs from Nutrient Flow in an ecosystem. [04]
(b) Enlist types of food chains in an ecosystem. Explain Ecological Pyramid of Energy. [05]
(c) Define 'Biogeochemical Cycling'. With a neat sketch explain carbon cycle in detail. [05]
- Q - 6 (a) Give detailed classification of natural resources. [04]
(b) Enlist the components of marine ecosystem. Which are the problems due to overexploitation of energy resources? [05]
(c) 1. As per EPA Environment is defined as..... [05]
2. Environmental Lapse Rate means.....
3. Ecological factor is defined as.....
4. During 1st Stages of formation of Atmosphere gases formed are.....
5. Define 'Point Source Pollutant'.

OR

- Q - 6 (a) Explain advantages and limitations of non – conventional energy resources. [04]
(b) Enlist components of forest ecosystem. Write advantages of renewable resources. [05]
(c) 1. Environmental Science is defined as [05]
2. Meteors found in the sphere is
3. Define 'Qualitative Pollutant'
4. As per Greek word Ecology term is integration of
5. Define 'Biosphere'.
